Emberwatch: Machine Learning and Integer Programming for Placing Wildfire Detection Drones in British Columbia

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The Problem

Fire seasons are getting bigger

2.84M hectares burned in BC in 2023, the most on record

2,296 wildfires recorded in that single season

50% of BC wildfires since 2000 ended smaller than 0.1 hectares

43,724 wildfires recorded in BC from 2000 to 2025

- Most fires stay small, and a few become enormous.
- A docked drone can launch on its own when a fire is reported, but only if a dock is close enough.
- BC now uses drones on active fires. The open question is where permanent docks should go.

Introduction

Emberwatch asks where British Columbia should place docked wildfire detection drones so that as many new fires as possible are within a 30-minute flight. It combines a machine learning forecast of where fires start with an integer program that chooses the sites.

Predict	Place	Replay	Stress-test
ML model estimates fires per 10 km cell	Optimizer picks the best 30 sites	Real seasons run fire by fire	Record year, drought and weather

- Every decision uses data up to the end of 2021 and is scored on the real fires of 2022 to 2025, including where it fails.

Objectives

- **Predict where wildfires will start** in each 10 km cell of BC, one season ahead
- Choose the 30 drone base sites that put the most predicted fires within a 30-minute flight
- Test every decision on fire seasons the model never saw
- Measure what a network would really deliver once drones are busy or grounded by weather
- Find where the model fails, and test whether drought and weather data fix it

Key Metrics

Metric	Definition
Coverage	Share of real wildfires within 45 km of a base, a 30-minute flight
Replay coverage	Share with a drone overhead within 30 minutes, after busy drones and weather
Top 10% capture	Share of next season's fires inside the 10% of cells ranked riskiest
AUC	How well the ranking separates cells with and without a fire; 0.5 is chance
Hindsight ceiling	Coverage if sites were chosen knowing the real fires, the best possible

Key References

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Engineering Methodology

Data Assembly → ML Risk Model → Placement Optimizer → Season Replay → Stress Tests

Data

43,724 BC Wildfire Service wildfires, 2000 to 2025

10,649 land cells of 10 km covering BC

421 candidate sites: 255 weather stations, 166 towns

42 features per cell and season

- Burned-area perimeters, roads and biogeoclimatic zones from the BC Data Catalogue
- Monthly weather for 479 points from NASA POWER
- Only weather stations installed before 2022 count as candidates

Season Replay

- Every 2022 to 2025 wildfire dispatched in time order to the drone that can arrive first
- A drone is busy for its flight, 15 minutes overhead and a 40-minute recharge
- 15% of calls grounded by weather; 20 replays per setting

ML Risk Model

Gradient-boosted decision trees, LightGBM, with a Poisson objective because the target is a count. Each row is one cell in one season, built only from earlier seasons. Trained on 2010 to 2019, tuned on 2020 to 2021, tested on 2022 to 2025.

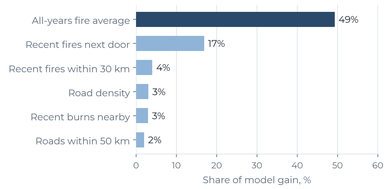


Figure 1. Share of the model's gain by feature. Fire history dominates; roads and recent burns add a little.

One season ahead, 2022 to 2025	Top 10%	AUC
10-season average	44.5%	0.790
Smoothed average	45.3%	0.837
ML risk model	**47.4%**	**0.849**

Placement Optimizer

A maximal covering location problem, written as an integer program and solved exactly with the CBC solver. Every run reached a proven optimum.

maximize $\Sigma_i\, w_i\, z_i$

subject to $z_i \le \Sigma_{j \text{ within } 45\text{ km}} y_j,\ \Sigma_j\, y_j = 30,\ y_j \in \{0, 1\}$

w_i = predicted fires in cell i; $y_j = 1$ if a base is built at site j

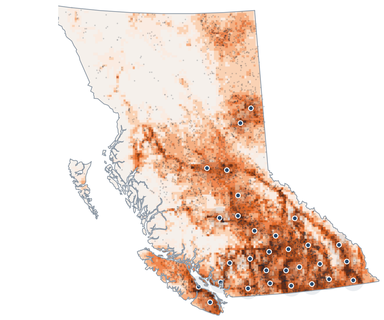
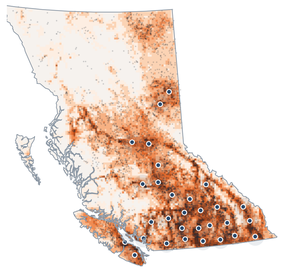


Figure 2. The 30 sites chosen with data up to 2021, over the predicted risk surface and the 2022 to 2025 wildfires.

Results

Optimized: 53% of fires within 30 min



Evenly spaced: 30% of fires within 30 min

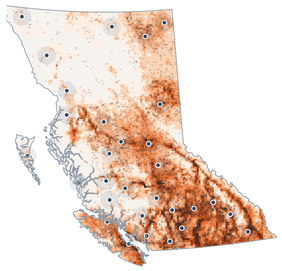


Figure 3. The same 30 bases placed by the optimizer and spaced evenly, scored on 7,192 real wildfires from 2022 to 2025.

Siting strategy	10 bases	20	30	60
ML risk model + optimizer	**23.8%**	**41.0%**	**52.8%**	**76.7%**
History average + optimizer	26.2%	40.9%	53.0%	76.5%
Evenly spaced	8.2%	19.8%	29.9%	53.3%
Random weather stations	11.0%	20.8%	28.8%	48.3%
Biggest towns	7.9%	11.2%	16.7%	33.7%
Hindsight ceiling	*26.9%*	*42.9%*	*55.2%*	*78.5%*

Table 1. Share of 2022 to 2025 wildfires within a 30-minute flight of a base.

Key finding: with 30 bases, optimized sites reach **76% more fires** within 30 minutes than evenly spaced bases, 52.8% against 29.9%. The gap held in every test year and at reaches of 30, 45 and 60 km.

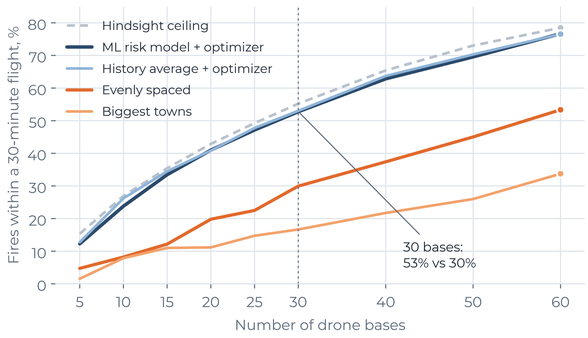


Figure 4. Coverage by number of bases. The ML model and plain history pick almost equally good sites, both within 3 points of hindsight, because a 45 km reach averages out cell-level differences.

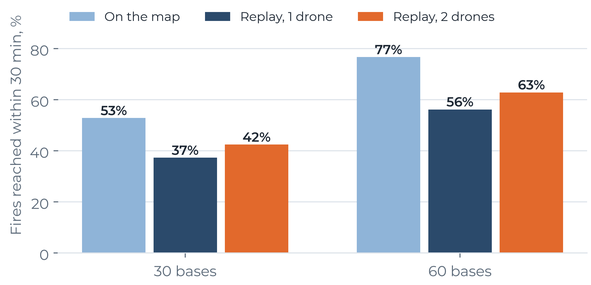


Figure 5. Coverage on the map against the fire-by-fire replay. Busy drones and weather cut 53% to 37%; a second drone per base raises it to 42%. On 22 July 2024, 139 fires started and one drone per base reached about 16% in time.

2023 Stress Test

2023 was BC's record season, so the model was retrained as if it were 31 December 2022 and then scored on what happened. It forecast **1,449** wildfires; there were **2,296**. It was close in the south and far off in the north, where fires had been rare.

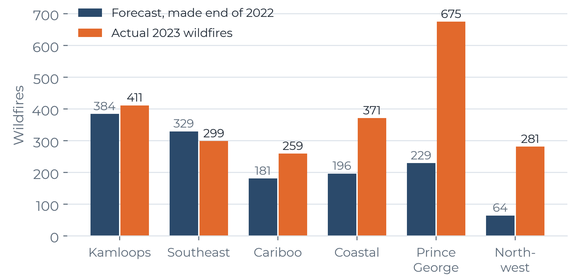


Figure 6. 2023 forecast against actual wildfires by BC Wildfire Service fire centre.

Notable 2023 fire	To base	Reached in 30 min
McDougall Creek, West Kelowna	5 km	20 of 20 replays
Bush Creek East, Shuswap	28 km	6 of 20 replays
Donnie Creek, largest in BC history	182 km	0 of 20 replays

Table 2. Bush Creek East started on a day with 55 new fires, so drones were often busy elsewhere.

Drought and Weather

To try to fix the 2023 miss, monthly temperature, rain, soil moisture and snow for 479 points were added, each as an anomaly from its own 1999 to 2019 normal.

Test, 2022 to 2025	Result
Drought known by 1 May, yearly forecast	47.9% vs 47.4%
2023 total, with spring drought data	still 46% low
Month-ahead regional error, weather added	51.3% → 47.8%
Bases moved monthly, May to Sep coverage	**51.2% → 55.7%**

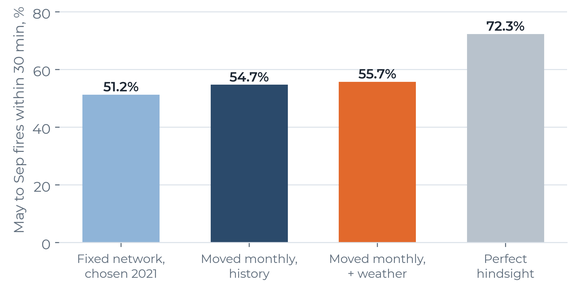


Figure 7. Coverage of 6,334 May to September wildfires when the 30 bases are re-sited on the 1st of each month.

Conclusion

- Siting bases on fire risk reached 76% more fires than even spacing, in every test year.
- The ML model ranked cells better than history but chose equally good sites.
- Busy drones and weather cut coverage by nearly a third; a few storm days set the capacity needed.
- Pre-season drought does not predict BC fire seasons. Flexible drone placement matters more than a better spring forecast.

Future Work

- **Technical report** with Prof. Xiaotao Bi, UBC, in preparation for the BC Wildfire Service
- Replace assumed drone specs with a real long-range drone's range and speed
- Tie weather groundings to recorded lightning, and add dock and drone costs
- Score the 2026 season once its data are released